

Alex Chen

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SUMMARY

Passionate Computer Science undergraduate with hands-on experience in machine learning, Python backend development, and scalable cloud systems. Experienced in building recommendation engines, deep learning computer vision models, and RESTful APIs using FastAPI and PyTorch.

EDUCATION

Bachelor of Technology (B.Tech) in Computer Science and Engineering

California Institute of Technology | Expected Graduation: 2025

GPA: 3.85 / 4.0

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Database Management Systems, Distributed Systems, Computer Networks.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C++, SQL

AI & Machine Learning: Machine Learning, Deep Learning, PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, OpenCV, NLP

Web & Backend: FastAPI, Django, React, REST API, Node.js, GraphQL

Databases & Cloud: PostgreSQL, MongoDB, Redis, Docker, Kubernetes, AWS, Git, GitHub, Linux, CI/CD

PROJECTS

Explainable Recommendation Engine

- Architected a hybrid job recommendation system using Sentence-Transformers (MiniLM) and cosine similarity.
- Developed multi-factor scoring combining semantic similarity, normalized skill overlap, and candidate eligibility.
- Built interactive frontend with React and TypeScript, delivering explainable match justifications for students.

Computer Vision Defect Detection Pipeline

- Implemented convolutional neural network in PyTorch and OpenCV achieving 94% precision on industrial inspection datasets.
- Optimized model inference latency by 35% using ONNX runtime and deployed containerized REST API with FastAPI and Docker.

Distributed Task Queue & Cache

- Built high-throughput asynchronous job processor using Python, Redis, and Celery processing 10k+ events/min.
- Integrated PostgreSQL for transactional persistence with SQLAlchemy ORM and Alembic migrations.

EXPERIENCE

Machine Learning Research Assistant

AI & Robotics Lab | June 2024 - Present

- Conducted experiments on self-supervised representation learning using PyTorch and HuggingFace transformers.
- Processed and cleaned large unstructured text and image datasets using Pandas and NumPy.

Software Engineering Intern

TechNova Systems | January 2024 - May 2024

- Contributed to microservices development using FastAPI, PostgreSQL, and Docker.
- Implemented CI/CD pipelines using GitHub Actions to automate unit testing and container deployment.

CERTIFICATIONS

- AWS Certified Cloud Practitioner
- DeepLearning.AI Deep Learning Specialization